Mastering Keycloak

Open Source Identity & Access Management

Course Introduction

Identity Simplified

Securing modern applications shouldn't be a hurdle.

What is Keycloak?

Keycloak is a powerful, open-source **Identity and Access Management (IAM)** solution targeted towards modern applications and services.

Maintained by Red Hat, it abstracts away the complex security logic, allowing developers to easily secure applications and APIs with little to no code.

Instead of building authentication from scratch, Keycloak provides a ready-to-use, robust authentication server.

Standard Protocols

OpenID Connect

The modern standard for authentication. Keycloak acts as an OIDC provider, seamlessly handling user identity and profile information across your ecosystem.

OAuth 2.0

The industry-standard protocol for authorization. Issue access tokens securely to allow third-party apps or your own frontends to access backend APIs.

SAML 2.0

Full support for enterprise-grade SAML 2.0, allowing easy integration with legacy corporate systems and external identity providers.

Key Features Overview

- **Single-Sign On (SSO):** Users authenticate once and can navigate across multiple applications seamlessly without re-entering credentials.
- **Identity Brokering:** Connect and authenticate users via external networks like Active Directory, or other OpenID Connect/SAML providers.
- **Social Login:** Enable quick onboarding with built-in support for social networks like Google, GitHub, Facebook, and Twitter.
- **Admin Console:** A centralized, intuitive web interface to manage realms, users, roles, sessions, and connected clients.

Why Choose Keycloak?

Building Custom Auth

- ✗ High development overhead & time
- ✗ Constant security maintenance required
- ✗ Implementing standard protocols is complex
- ✗ Highly prone to security vulnerabilities

Deploying Keycloak

- ✓ Out-of-the-box, ready-to-use solution
- ✓ Maintained by a massive open-source community
- ✓ Strictly compliant with OIDC & OAuth2
- ✓ Regular security patches and updates

By The Numbers

100%

Free & Open Source

Zero Licensing Fees

Keycloak offers enterprise-grade security features without the enterprise price tag. It empowers developers and architects to deploy highly scalable, secure identity solutions without restrictive vendor lock-in or expensive subscription costs.

Integration Scenarios

Integrate Keycloak seamlessly across your entire application stack.

Whether you are securing modern Single-Page Applications (SPAs), native mobile applications, or distributed backend microservices, Keycloak provides robust adapters.

We will explore how to secure varied architectures effortlessly within this course.

Course Roadmap

1. Core Concepts

Understanding Realms, Users, Roles, and the Admin interface.

2. Securing Apps

Integrating SPAs and APIs using OIDC and OAuth 2.0 flows.

3. Advanced Config

Implementing Identity Brokering, Social Login, and Federation.

4. Production

Custom theming, deployment best practices, and performance.

Ready to Begin?

Enroll now and start securing your applications.

Image Sources

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